# Dominion Energy Virginia reliability: what public data shows, and where I'd look first

Independent analysis of public data

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November 3, 2026

Data: EIA-861 · PUDL · NOAA Storm Events · EAGLE-I · EIA-930

# Outage minutes rose from 130 to 133; 2025 had more storms and more interruptions

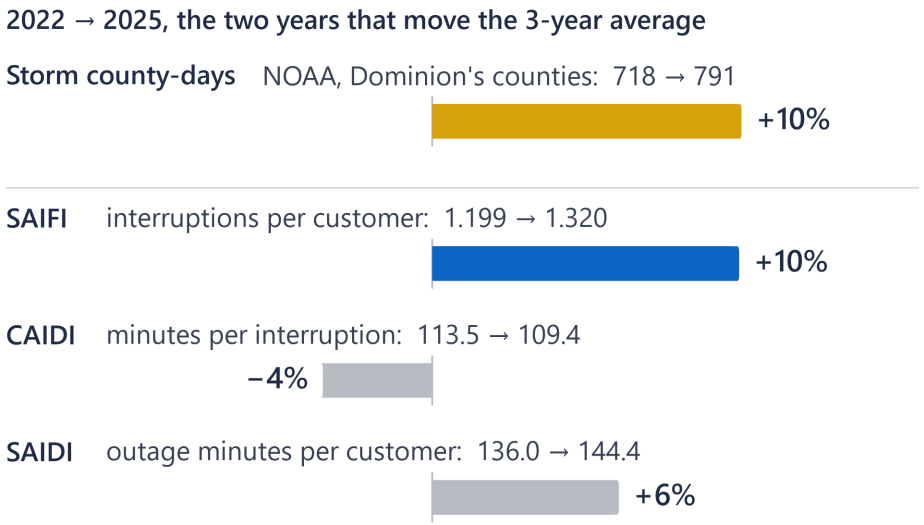
## 130 → 133

Reported 3-year average outage minutes per customer, excluding major events

**Rebuilt from EIA-861: 130.2 → 133.0**

2025 (144.4) replaced 2022 (136.0) in the window.

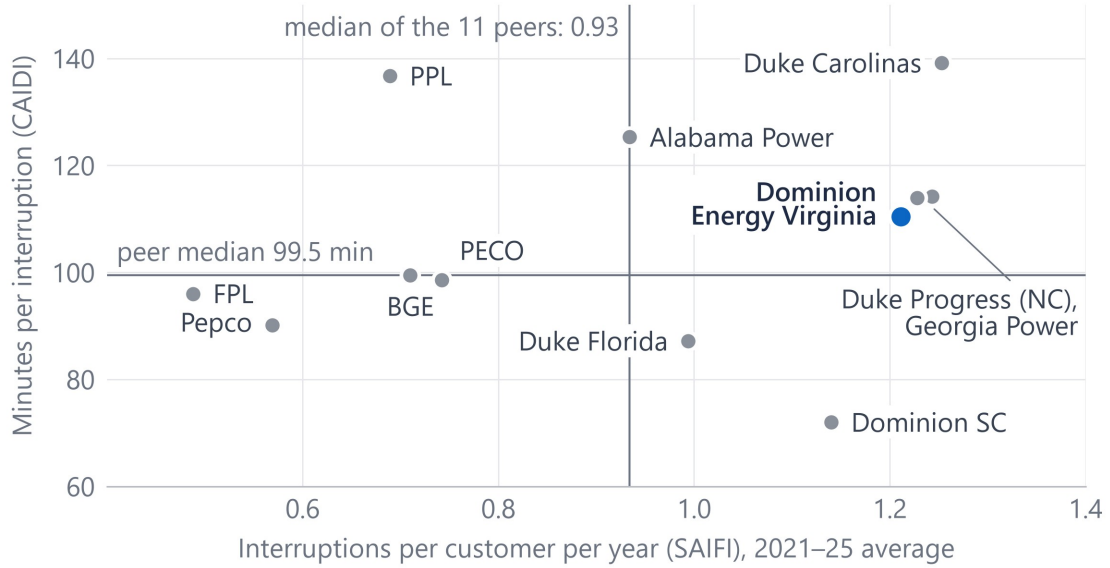
Dominion attributes the rise primarily to increased storm activity (2025 Annual Report, p. 10). 2025 sits on the storm line, with no clear trend once storms are counted (A2).




**Why it matters:** a stormy year can move a 3-year average. With outage records, I'd ask whether non-storm interruptions are rising underneath.

**Source:** EIA-861 2025 final and PUDL (EIA-861) 2013–2024, Dominion Energy Virginia, excluding major event days; 3-year means of annual values. NOAA Storm Events (A2). Reported figures: Dominion Energy 2025 Annual Report, p. 10.

# Among 12 large utilities, Dominion's interruption length is mid-pack; frequency is the gap




**1.21**

interruptions per customer a year: 4th highest of the 12 (peer median 0.93)

**110 min**

per interruption: mid-pack, 6th of the 12 (peer median 99.5)

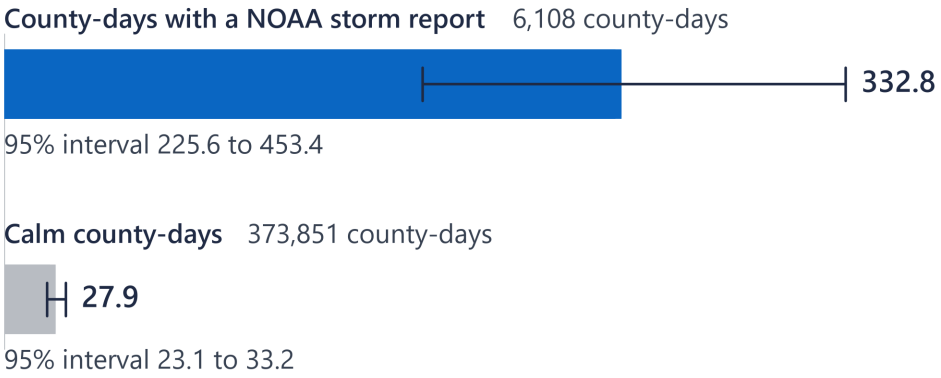
Methods differ: Dominion uses its own (non-IEEE) major-event rule, most peers IEEE 1366, so ranks are indicative.

**Why it matters:** if frequency is where Dominion differs, the next question is which causes drive interruptions, and that needs outage records.

**Source:** EIA-861 / PUDL 2021–2025, excluding major event days, means of annual values. Dominion and 11 peers: twelve large investor-owned utilities in PJM and the Southeast, selected by EIA ID before results were computed; medians exclude Dominion.

# County-days with storms had about 12 times the outage hours of calm days

**Outage customer-hours per 1,000 customers, mean per county-day, 2015–2025**




11.9×

storm vs calm county-days (95% CI 7.9–16.9)

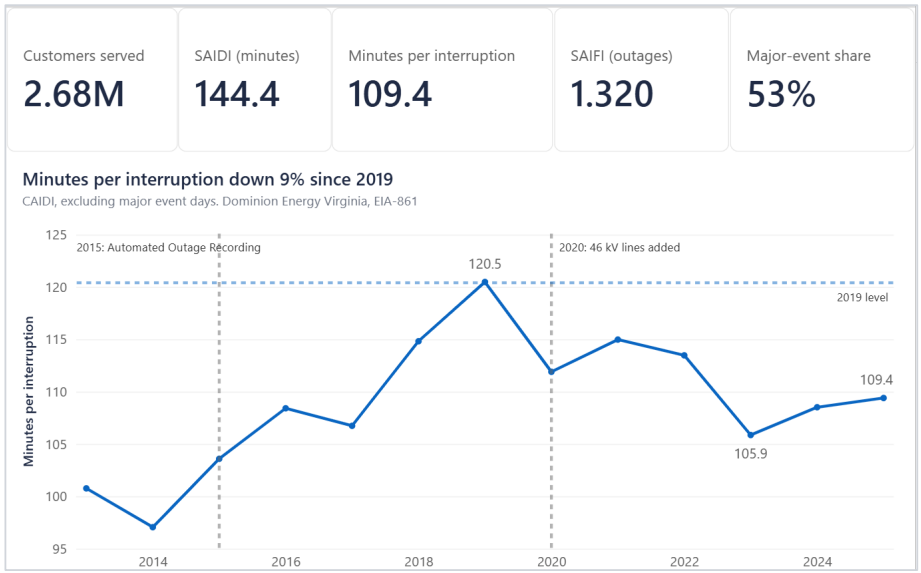
Holds on non-major days (9.9×), in Dominion-only counties (19.9×) and in every year 2015–2025.

EAGLE-I counts every utility's customers in a county. An association: storm reports partly follow damage.

**Why it matters:** reporting reliability next to storm exposure would keep a stormy year from being read as a trend.

**Source:** EAGLE-I (ORNL) and NOAA Storm Events, 2015–2025: 6,108 storm county-days with complete outage readings in 99 counties (EAGLE-I); complete days only (2025: 74% of county-days). 95% intervals: 2,000 resamples of whole days.

# The Power BI report: one page per question, on customer-weighted measures




A detail of the Summary page: the 2025 cards and the CAIDI trend

**FCT_RELIABILITY**
utility × state × year

many:1

**DIM_UTILITY**
peer group

A small star at the core; the other tables are pre-built summaries for particular pages, so most stand alone. 18 tables, one DataFolder parameter.

**Customer-weighted measures; the cards follow the year slicer:**

```
Selected SAIDI =
VAR SelectedYear =
    SELECTEDVALUE('Dashboard Year'[Year], 2025)
RETURN
    CALCULATE([SAIDI ex MED],
        FCT_RELIABILITY[YEAR] = SelectedYear)
```

**Why it matters:** Power BI reporting leads the role's duties, and the same model and measures could carry a program-tracking report.

**Source:** Power BI Desktop, the project's report: a detail of the Summary page, year 2025. Tables are CSV exports (16 from the local DuckDB mirror, 2 from the workbook) imported through a folder parameter; measures as in the project's measures.dax.

# Built with production habits: tested, documented, and reconciled to the reported numbers

Original build, Oct 27–31, 2026 | Added Nov 1, 2026

| Sources | Downloads | SQL layers | Tests | CSV exports | Power BI Import |
|---|---|---|---|---|---|
| EIA-861 / PUDL, NOAA Storm Events, EIA-930 | download scripts | sql/01–08: RAW → STG → MART | T1–T12 | | |
| EAGLE-I, DOE-417, HURDAT2, degree days, Census, EIA-861M | scripts for the new sources, each with a manifest | sql/09–11; all of it run in a local DuckDB mirror (22 s) | T13–T31: all 31 pass | 18 files with a manifest of row counts and checksums | template: 18 tables via one DataFolder setting; opens offline |

**Snowflake:** holds the October build, sql/01–08 and T1–T12, where the report was developed in DirectQuery (D13). The same SQL loads the new tables there in one optional step (README step 7).

**Reconciled:** 12 of 12 match the published figures.

**Documented:** data dictionary; decisions D1–D22; peers fixed by EIA ID before results.

**Why it matters:** the same habits would serve the team's reports: reconcile first, test every load, write each definition down.

**Source:** Project repository: sql/01–11, tests T1–T31 (T1–T12 the October build), data dictionary, decisions log D1–D22. Build time of the local DuckDB mirror measured November 1, 2026.

# With Dominion's outage, meter and circuit data: the questions I'd bring and what I'd build

Days 1–30

## Learn and reconcile

Learn how outages, meter events and circuits are recorded and reported.

Rebuild one year's SAIDI, SAIFI and CAIDI and reconcile them to the filed figures.

**Deliverable:** a reconciled baseline with written definitions

Days 31–60

## Ask about frequency

By circuit and cause code, storm days apart from calm days; repeat interruptions (CEMIn).

Outage times from meter last-gasp and restore messages (99.8% of meters are smart), checked against OMS.

**Deliverable:** tested views and a data dictionary

Days 61–90

## Build the tracking report

Track mainfeeder hardening against matched untreated feeders (same pre-period performance), weather-normalized, in support of the program's selection rule.

**Deliverable: a Power BI program-tracking report, refreshed on a schedule**

**Why it matters:** these are questions to check first with the people who own each definition, not conclusions.

**Source:** A plan: Dominion data is named by type only, and none was used here. Smart meters: EIA-861 2025 (advanced meters as a share of all meters). Grid Resiliency: Dominion Energy website, read November 1, 2026.

# Three takeaways: a stormier year, a frequency gap, and storms by county

1 **A stormier year**
More storms and more interruptions in 2025; no clear trend once storms are counted

2 **Frequency is the peer gap**
So causes are the question, and answering it needs outage records by cause code

3 **Storms by county**
Storm county-days carry far more outage hours, so report reliability beside storm exposure

## I've built a tested Snowflake → Power BI pipeline on public reliability data.

**How my experience maps** Mercury Insurance, Data Analyst

**Snowflake pipeline, 20M+ rows**
→ this project's mart

**Root-cause analysis on joined data**
→ what I'd do with cause codes in the first 90 days

**Metadata and business definitions**
→ the data dictionary

**Thank you. I'd like to do this work with your team.**

**Source:** Takeaways: slides 2–4. Experience: Mercury Insurance, Data Analyst, 2026 to present.

# Definitions: SAIDI = SAIFI × CAIDI, and major event days are set aside by rule

**SAIDI**
outage minutes per customer per year

=

**SAIFI**
interruptions per customer per year: how often

×

**CAIDI**
minutes per interruption: how long

**Dominion 2025:** 1.320 × 109.4 = 144.4 minutes (excluding major event days)

Major event days (MED): IEEE 1366's 2.5 beta method takes five years of daily SAIDI, drops zero days, and sets α and β as the mean and standard deviation of ln(daily SAIDI); a day above exp(α + 2.5β) is an MED.

Dominion reports to EIA under a non-IEEE rule ("Other" in EIA-861); most peers in this deck use IEEE 1366.

With major event days, 2025 SAIDI was 306.7 minutes, against 144.4 without them: MEDs held 53%.

Log split: ln(SAIDI ratio) = ln(SAIFI ratio) + ln(CAIDI ratio). From 2022, duration fell, so frequency accounts for more than all of the rise; from 2023, 79%.

**Source:** IEEE Std 1366; EIA-861 reliability, Dominion Energy Virginia, 2022, 2023 and 2025.

# The weather fit is a check on one year, not a forecast

| Year | Storm county-days | SAIDI ex-MED | From the fit | Difference |
|---|---|---|---|---|
| 2015 | 521 | 119.6 | 131.3 | −11.6 |
| 2016 | 594 | 138.2 | 135.0 | +3.2 |
| 2017 | 419 | 117.4 | 126.0 | −8.6 |
| 2018 | 683 | 149.6 | 139.7 | +9.9 |
| 2019 | 848 | 149.9 | 148.2 | +1.7 |
| 2020 | 429 | 141.0 | 126.5 | +14.5 |
| 2021 | 673 | 133.6 | 139.1 | −5.5 |
| 2022 | 718 | 136.0 | 141.5 | −5.4 |
| 2023 | 390 | 123.3 | 124.5 | −1.1 |
| 2024 | 464 | 131.3 | 128.3 | +3.0 |
| **2025 (not in fit)** | **791** | **144.4** | **145.3** | **−0.9** |

Fit on 2015–2024 (n = 10): SAIDI = 104.3 + 0.0518 × storm county-days; r = 0.70 (t = 2.75, p = 0.025; 95% CI 0.12 to 0.92), $R^2$ 0.49.

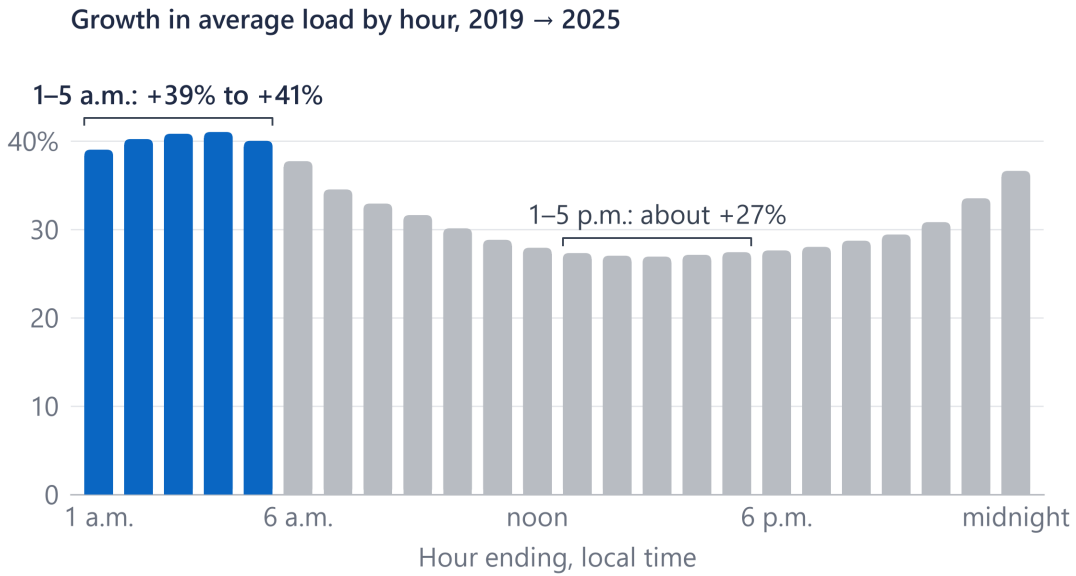
Why 2015: the change to automated outage recording that year shifts the level (decision D6).

2025, left out of the fit: 145.3 from the fit (95% prediction interval 122.3–168.2), 144.4 actual.

Adjusted for storm days, the trend since 2015 is not distinguishable from zero: about +0.3 minutes a year (95% range −1.4 to +2.1).

**Source:** NOAA Storm Events and EIA-861, 2015–2025; ordinary least squares on ten annual points (exploratory). Storm county-days count Bedford city once (decision D22).

# Overnight demand grew faster than afternoon demand; winter rivals summer for the peak




## +39–41%

average load, hours ending 1–5 a.m., 2019 → 2025 (1–5 p.m.: about +27%)

## Winter

set the yearly peak in 2019, 2022 and 2025; 2026 so far: winter 25,413 vs summer 25,384 MW, a tie

Highest hour of the year +26%, lowest hour +52%.

**I'd add winter and overnight views to load and outage reports.**

**Source:** EIA-930 hourly demand, PJM Dominion (DOM) zone, which is wider than Dominion's retail territory; 2019 and 2025 full years; 2026 to the hour ending midnight, Oct 29. Winter = Dec–Feb, summer = Jun–Aug.

# Every number traces to a public source; the tables behind them are tested after each load

| Source | Publisher | Years | Grain | Rows |
|---|---|---|---|---|
| EIA-861 Reliability, 2025 final | EIA | 2025 | utility × state × year | 977 |
| PUDL EIA-861 reliability | Catalyst Cooperative | 2013–2024 | utility × state × year | 24,358 |
| EIA-861 Service Territory | EIA | 2025 | utility × county | 11,803 |
| NOAA Storm Events | NOAA NCEI | 2013–2025 | storm event | 831,288 |
| EAGLE-I county outages | ORNL for DOE | Nov 2014–2025 | county × 15 minutes | 11,608,442 |
| EIA-930 hourly demand | EIA | Jul 2018–Oct 2026 | region × hour | 6,094,320 |
| HURDAT2 storm tracks | NOAA NHC | 1851–2025 | storm × 6-hour fix | 55,524 |
| DOE-417 disturbance reports | DOE | 2014–2025 | reported event | 3,702 |
| CPC degree days | NOAA CPC | 2018–Oct 2026 | state × day | 309,504 |
| Dominion Energy 2025 Annual Report | Dominion Energy | 2025 | document (130, 133) | reference |

**Tests T1–T31, all pass:** T1–T6 on the reliability mart (no duplicate keys; SAIDI = SAIFI × CAIDI within 2%; Dominion 2025 equals EIA's published 306.7, 144.4 and 2,675,735 customers); T7–T9 on NOAA storms; T10–T12 on EIA-930 demand; T13–T31 on the new tables. **Reconciled 12 of 12** against the annual report and the EIA-861 files.

**Source:** Project documentation: sources register (publisher, vintage, license) and data-quality tests; all public data.

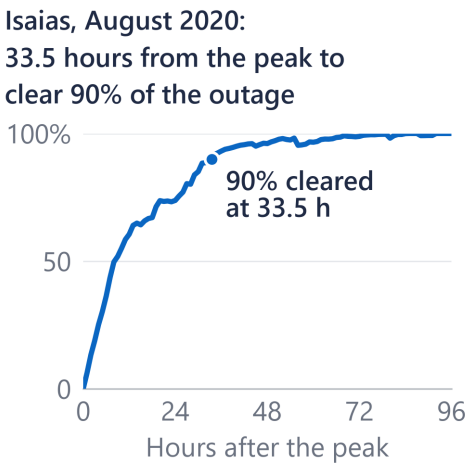
# What public data cannot show, and what would answer it

| Public data limit | What would answer it (Dominion data, by type) |
|---|---|
| One number per utility per year (EIA-861) | Outage records by event, circuit and cause |
| Storms counted by county, not by line | Outage records with cause codes and circuit location |
| Methods differ between utilities | Dominion's daily SAIDI and its filing method |
| The DOM zone is wider than the retail territory | Meter interval data by circuit |
| No customer identities: growth can't be attributed | Customer class and large-load records |

**Source:** Project scope limits (decision D14) and the sources above.

# The ten largest outage events since 2014: three tropical, two winter, five wind

| # | Peak day | Storm type | Peak customers out | Held 1 h | Hours to clear 90% |
|---|---|---|---|---|---|
| 1 | Oct 12, 2018 | Tropical (Michael) | 558,729 | 545,548 | 60.5 |
| 2 | Mar 2, 2018 | Wind | 419,136 | 401,776 | 55.0 |
| **3** | **Aug 4, 2020** | **Tropical (Isaias)** | **351,738** | **343,492** | **33.5** |
| 4 | Jan 3, 2022 | Winter | 314,644 | 308,015 | 102.8 |
| 5 | Feb 13, 2021 | Winter | 281,707 | 271,359 | 101.5 |
| 6 | Oct 9, 2016 | Tropical (Matthew) | 269,298 | 265,401 | 52.8 |
| 7 | Jun 16, 2016 | Wind | 234,951 | 233,003 | 65.8 |
| 8 | Jun 19, 2025 | Wind | 225,609 | 209,475 | 22.0 |
| 9 | Feb 16, 2025 | Wind | 187,074 | 172,538 | 20.3 |
| 10 | May 14, 2018 | Wind | 159,532 | 151,385 | 20.0 |




**Source:** EAGLE-I (ORNL), all utilities in Dominion's 103 counties, Nov 2014–2025; storm type from NOAA and HURDAT2. Cleared = (peak − customers out) ÷ (peak − pre-event level): a net count, not a utility's "% of affected customers restored".